Teorie

AVL ⟶ mai multe accesări decât B

1. Care din urmatorele argumente este esential pentru a alege un arbore B in loc de un arbore AVL pentru indexarea unor baze de date.

Select one:
- a. datele se preiau din memoria externa in blocuri
- b. inregistrarile in bazele de date sunt deja sortate dupa o cheie primara
- c. bazele de date au un numar mare de inregistrari
- d. arborii B necesita mai putina memorie decat arborii AVL

Clear my choice

multă

2. Un arbore B creste in inaltime doar atunci cand:

pagina nouă se face în sus, la scindare

Select one:
- a. se adauga un nou nod intr-o pagina terminala
- b. in niciuna din celelalte variante
- c. are loc o scindare a paginii radacina
- d. se adauga o noua pagina terminala

Clear my choice

3. Un arbore B isi reduce dimensiunea doar daca:

Select one:
- a. se sterge un nod dintr-o pagina terminala
- b. se face o echilibrare
- c. radacina este redusa la dimensiunea 0

Clear my choice

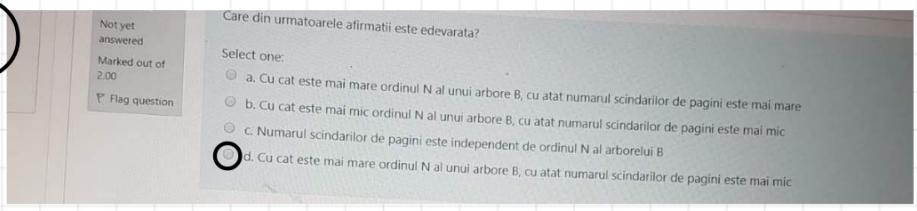
4. 1

Care din urmatorele argumente este esential pentru a alege un arbore B in loc de un arbore AVL pentru indexarea unor baze de date.

Select one:
- a. datele se preiau din memoria externa in blocuri
- b. inregistrarile in bazele de date sunt deja sortate dupa o cheie primara
- c. bazele de date au un numar mare de inregistrari
- d. arborii B necesita mai putina memorie decat arborii AVL

Clear my choice

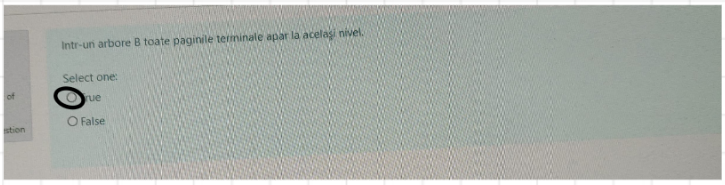
5.



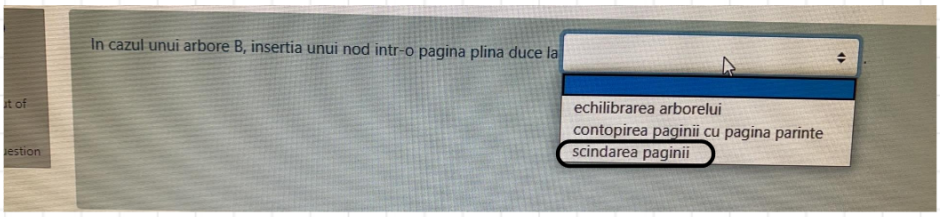
6. Lungimile drumurilor de la radacina la oricare nod terminal sunt egale intr-un arbore B.

Select one:
- True
- False

7.



8.



9.

Arbori B sunt adesea utilizati pentru

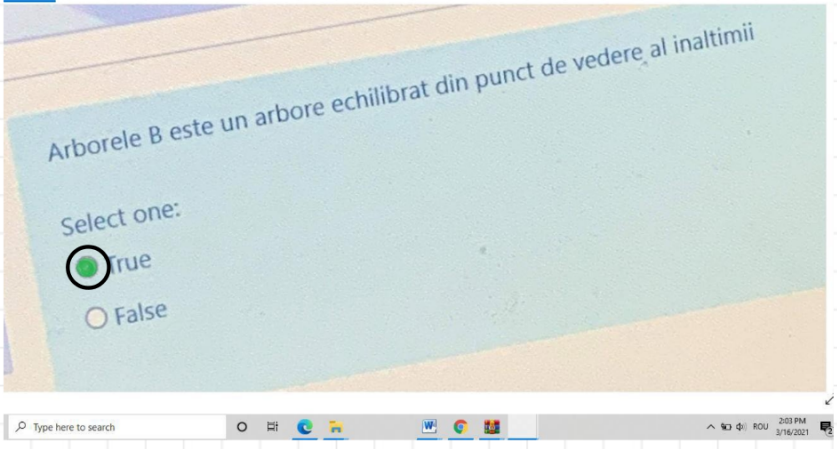
Select one:

- a. gestionarea unor date aflate in memoria externa
- b. gestionarea unor date fara a le retine intr-un mod ordonat dupa chei
- c. gestionarea unui numar redus de date
- d. implementarea dictionarelor de cuvinte

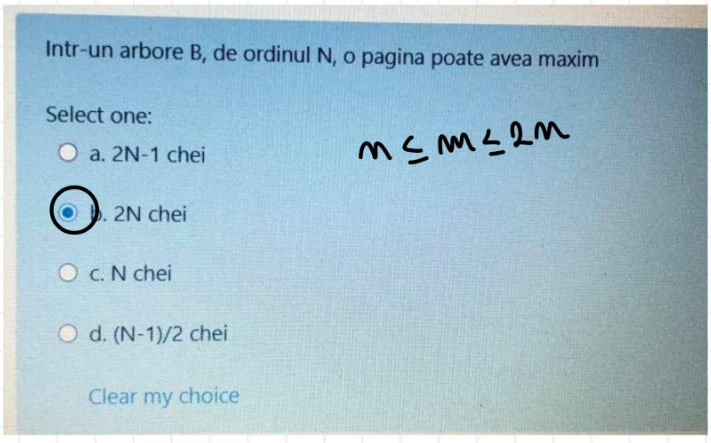
Huffman maybe

Clear my choice

10.



11.



$n \leq m \leq 2n$

Insertia

1.

Sa se scrie configuratia arborelui B cu N=1, dupa insertia urmatoarelor chei in ordinea data:

7,4,5,6,9,8,1,2

Nota:

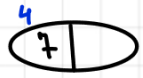
Arborele se va scrie in modul urmator: Fiecare pagina incepe cu paranteza patrata deschisa [si se termina cu cea inchisa], fiecare cheie va fi separata de cheia altui nod din aceeasi pagina, prin virgula (,), iar bara verticala | va separa nivelurile intre ele.

Exemplu:

[B,D]|[A][C]|[F,G] reprezinta un arbore B cu doua niveluri, avand pe primul nivel radacina cu cheile B si D si pe al doilea nivel trei pagini: prima pagina cu cheia A, a doua cu cheia C si a treia cu cheile F si G.

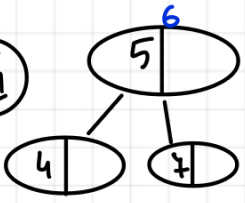
Answer: [5]|[2][7]|[1][4][6][8,9]

I

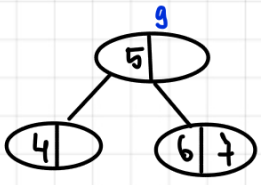


II

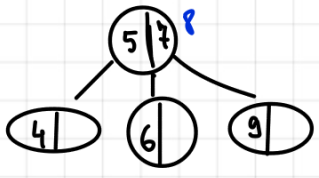
III



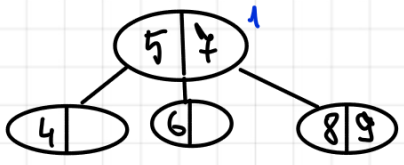
IV



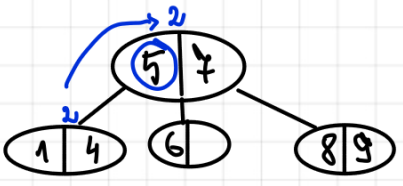
V.

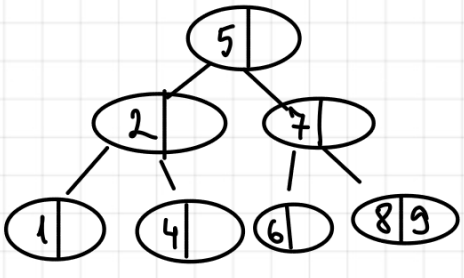


VI.



VII.





[5] | [2] [7] | [1] [4] [6] [8,9]

2.

Sa se scrie configuratia arborelui B cu N=1, dupa insertia urmatoarelor chei in ordinea data:

7,9,8,1,4,3,2,5

Nota:

Arborele se va scrie in modul urmator: Fiecare pagina incepe cu paranteza patrata deschisa [si se termina cu cea inchisa], fiecare cheie va fi separata de cheia altui nod din aceeasi pagina, prin virgula (,), iar bara verticala | va separa nivelurile intre ele.

Exemplu:

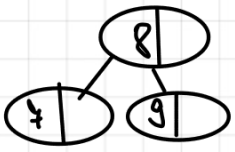
[B,D]|[A][C]|[F,G] reprezinta un arbore B cu doua niveluri, avand pe primul nivel radacina cu cheile B si D si pe al doilea nivel trei pagini: prima pagina cu cheia A, a doua cu cheia C si a treia cu cheile F si G.

Answer: [4] | [2][8] | [1][3][5,7][9]

I.

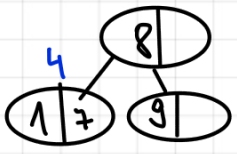
II.

III.



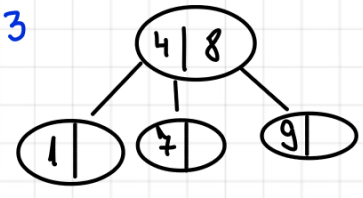
IV.

4



V

3



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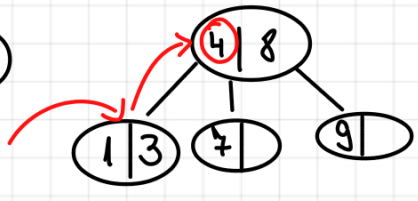
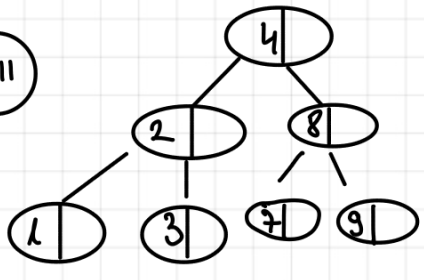


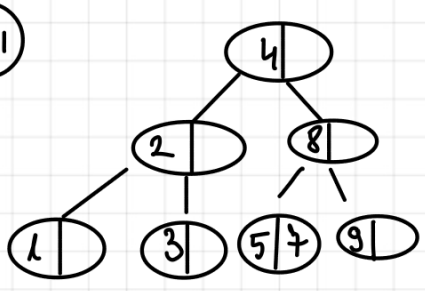
2

VII



5

VIII



3.

Sa se scrie configuratia arborelui B cu N=1, dupa insertia urmatoarelor chei in ordinea data:

2,1,3,6,7,4,5

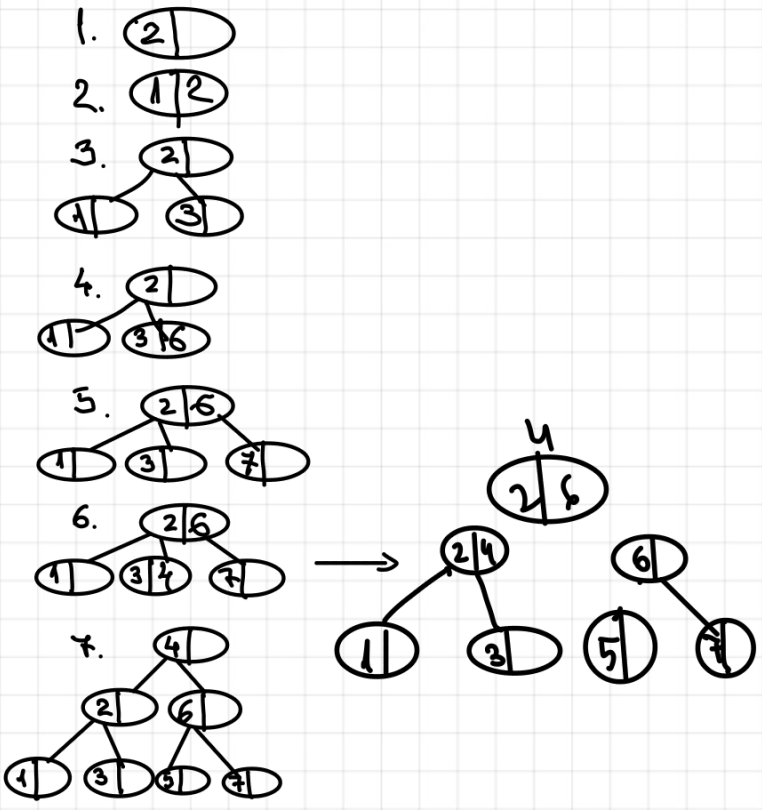
Nota:

Arborele se va scrie in modul urmator: Fiecare pagina incepe cu paranteza patrata deschisa [si se termina cu cea inchisa], fiecare cheie va fi separata de cheia altui nod din aceeasi pagina, prin virgula (,), iar bara verticala | va separa nivelurile intre ele.

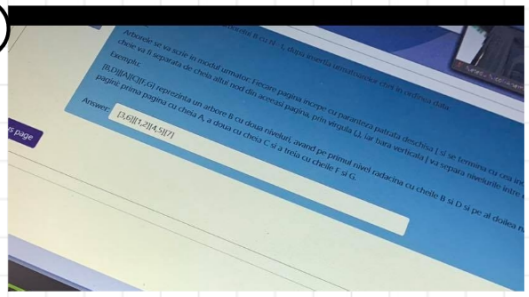
Exemplu:

[B,D]|[A][C][F,G] reprezinta un arbore B cu doua niveluri, avand pe primul nivel radacina cu cheile B si D si pe al doilea nivel trei pagini: prima pagina cu cheia A, a doua cu cheia C si a treia cu cheile F si G.

Answer: [4]|[2][6]|[1][3][5][7]



4.



5.

Sa se scrie configuratia arborelui B cu N=1, dupa insertia urmatoarelor chei in ordinea data:

5,3,2,1,4,7

Nota:

Arborele se va scrie in modul urmator: Fiecare pagina incepe cu paranteza patrata deschisa [si se termina cu cea inchisa], fiecare cheie va fi separata de cheia altui nod din aceeasi pagina, prin virgula (,), iar bara verticala | va separa nivelurile intre ele.

Exemplu:

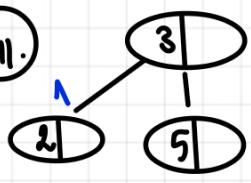
[B,D]|[A]|[C]|[F,G] reprezinta un arbore B cu doua niveluri, avand pe primul nivel radacina cu cheile B si D si pe al doilea nivel trei pagini: prima pagina cu cheia A, a doua cu cheia C si a treia cu cheile F si G.

Answer: [3,5]|[1,2]|[4]|[7]

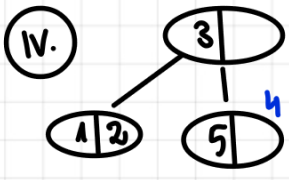
I.

II.

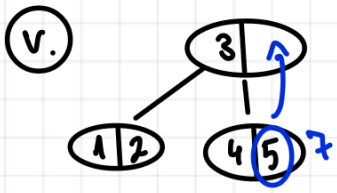
III.



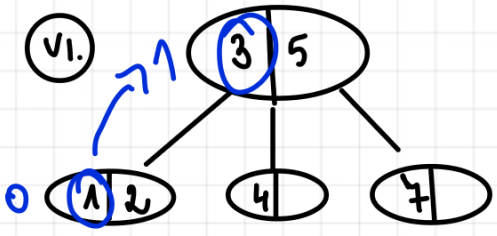
IV.



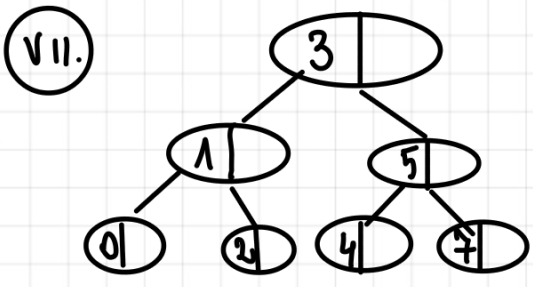
V.



VI.

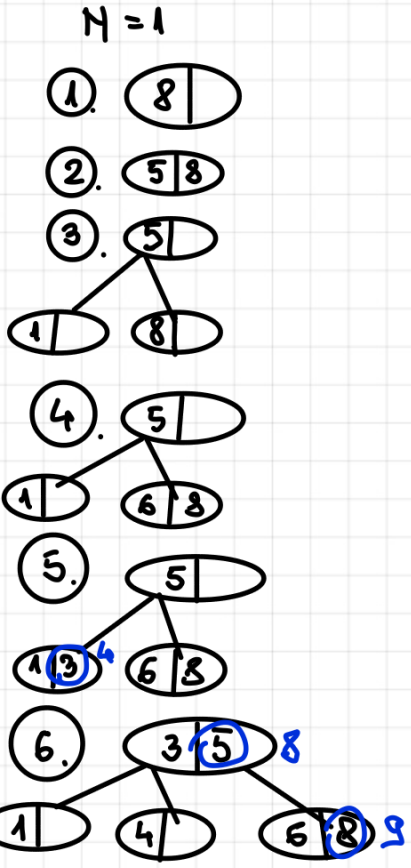


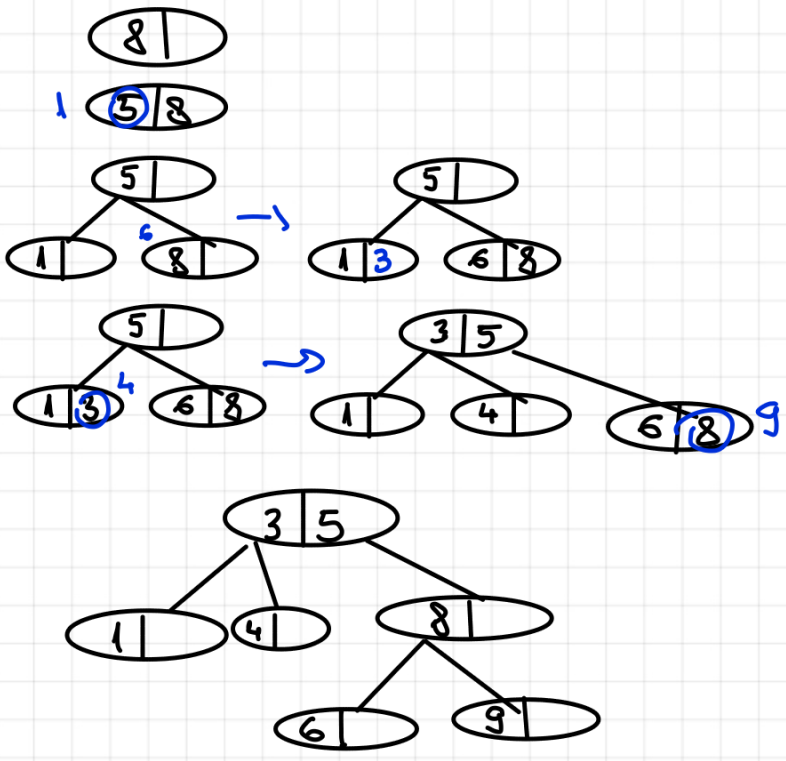
VII.

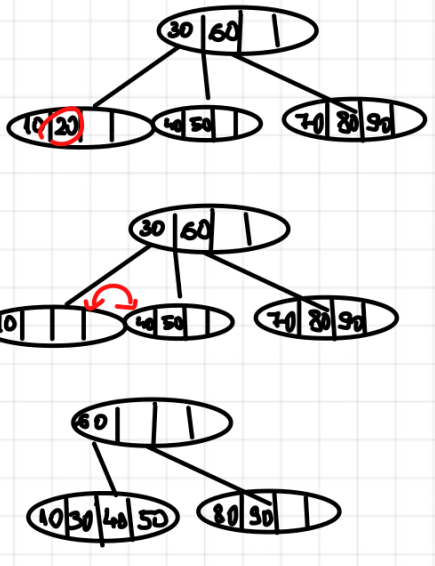


6. 8, 5, 1, 6, 3, 4, 9

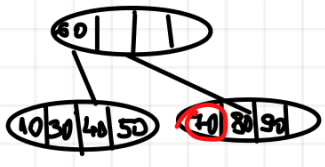
N = 1





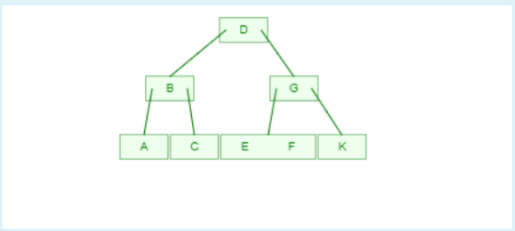


20, 60, 70



Suprimări

1. Sa se scrie configuratia arborelui B cu N=1 ⟶ $n \leq m \leq 2n$



avem max 2el

subdepășire → $m < n$

dupa suprimarea urmatoarelor chei in ordinea data:

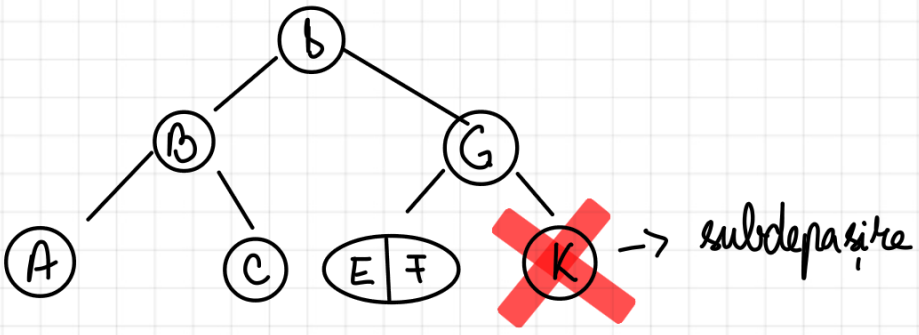
K,B

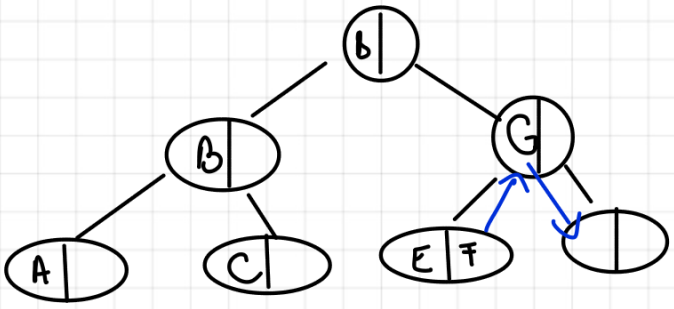
Nota:

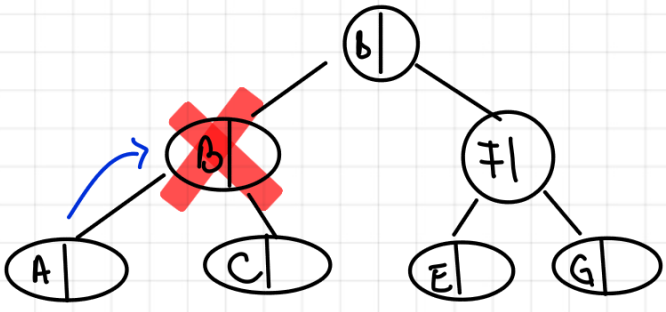
Arborele se va scrie in modul urmator: Fiecare pagina incepe cu paranteza patrata deschisa [si se termina cu cea inchisa], fiecare cheie va fi separata de cheia altui nod din aceeasi pagina, prin virgula (,), iar bara verticala | va separa nivelurile intre ele.

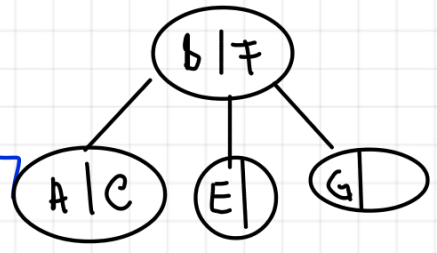
Exemplu:

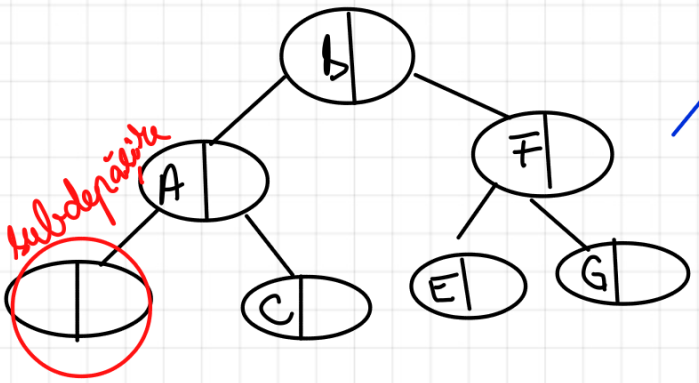
[B,D][A][C][F,G] reprezinta un arbore B cu doua niveluri, avand pe primul nivel radacina cu cheile B si D si pe al doilea nivel trei pagini: prima pagina cu cheia A, a doua cu cheia C si a treia cu cheile F si G.





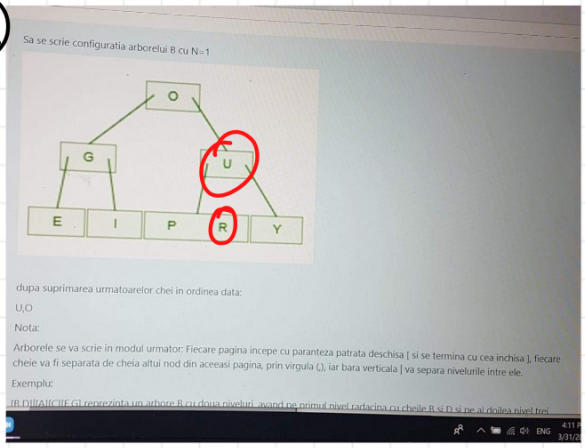


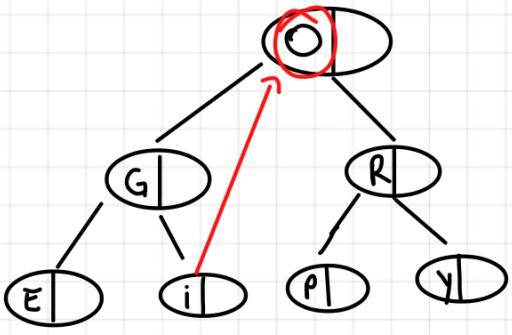


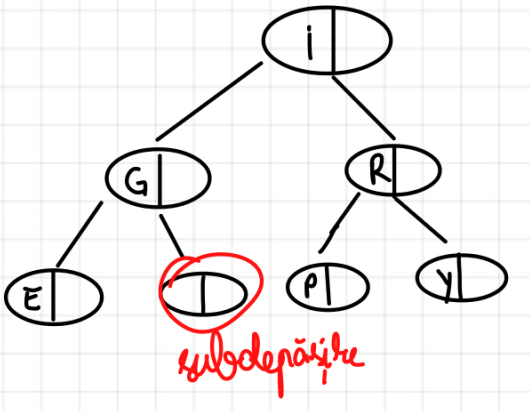


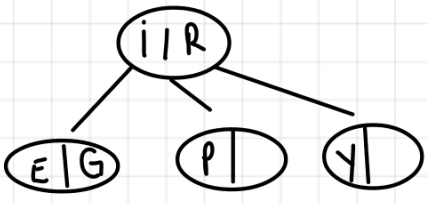
[b,f] | [A,C] [E] [G]

2.


Sa se scrie configuratia arborelui B cu N=1
O
G
U
E
I
P
R
Y
dupa suprimarea urmatoarelor chei in ordinea data:
U,O
Nota:
Arborele se va scrie in modul urmator: Fiecare pagina incepe cu paranteza patrata deschisa [si se termina cu cea inchisa], fiecare cheie va fi separata de cheia altui nod din aceeasi pagina, prin virgula (,), iar bara verticala | va separa nivelurile intre ele.
Exemplu:

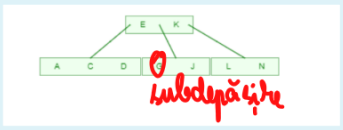

O
G
R
E
i
P
Y


i
G
R
E
P
Y
subdepășire


i | R
E | G
P
Y

3.

Sa se scrie configuratia arborelui B cu N=2



$2 \leq m \leq 4$

dupa suprimarea urmatoarelor chei in ordinea data:

G,N

Nota:

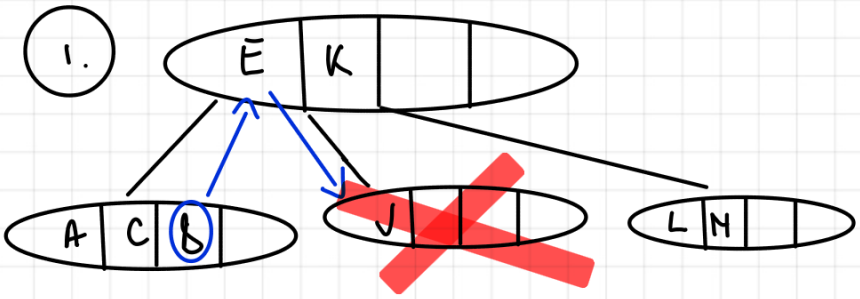
Arborele se va scrie in modul urmator: Fiecare pagina incepe cu paranteza patrata deschisa [si se termina cu cea inchisa], fiecare cheie va fi separata de cheia altui nod din aceeasi pagina, prin virgula (,), iar bara verticala | va separa nivelurile intre ele.

Exemplu:

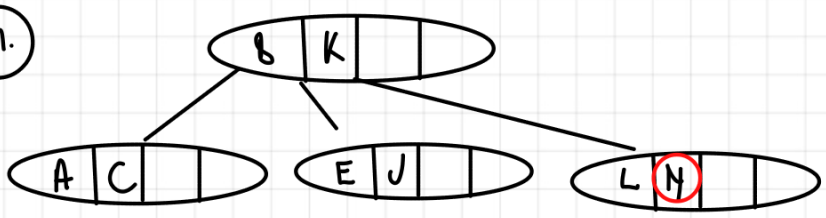
[B,D]|[A]|[C]|[F,G] reprezinta un arbore B cu doua niveluri, avand pe primul nivel radacina cu cheile B si D si pe al doilea nivel trei pagini: prima pagina cu cheia A, a doua cu cheia C si a treia cu cheile F si G.

Answer: [D]|[A,C]|[E,J,K,L]

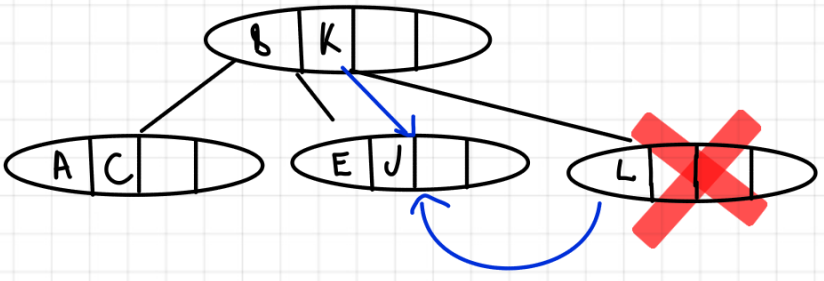
I.



II.

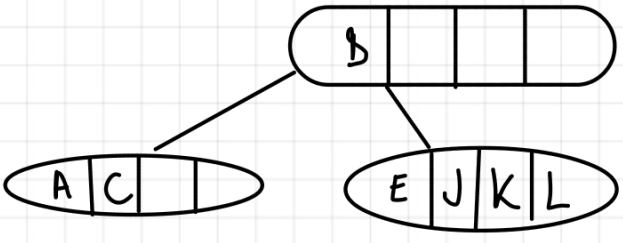


III.



subdepășite

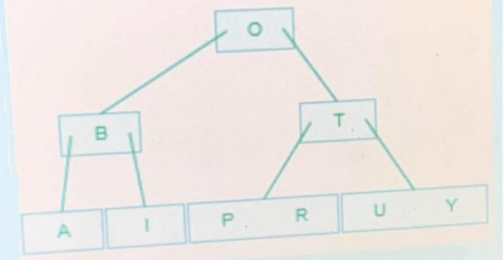
nu ne putem împrumuta



[B] | [A,C] [E,J,K,L]

4.

Sa se scrie configuratia arborelui B cu N=1

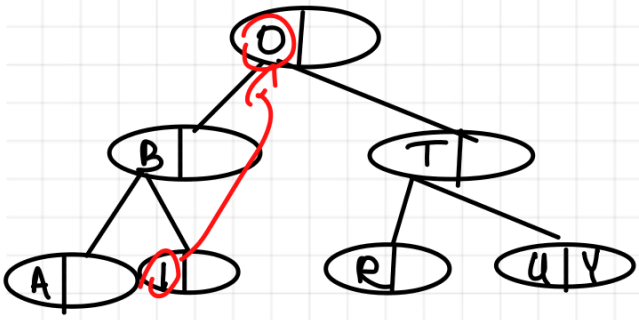


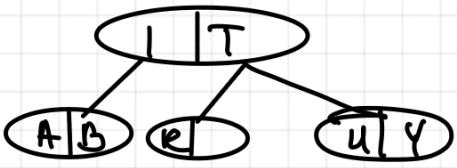
dupa suprimarea urmatoarelor chei in ordinea data:

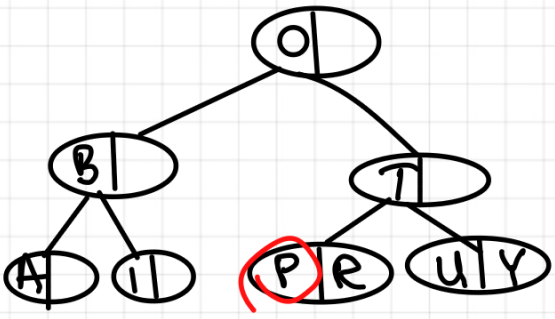
[I,T][A,B][R][U,Y]

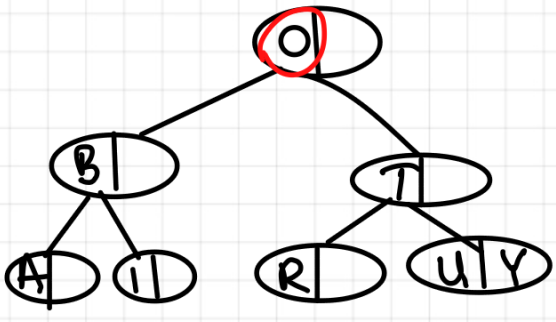
P,O

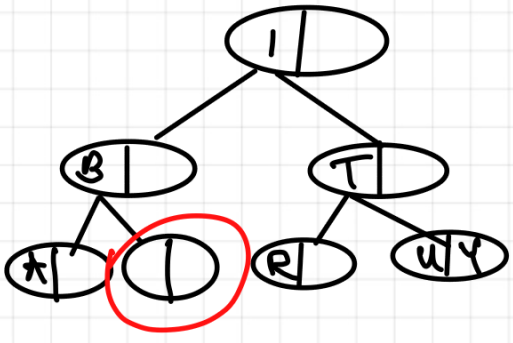
Nota:

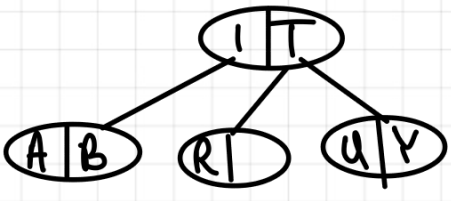




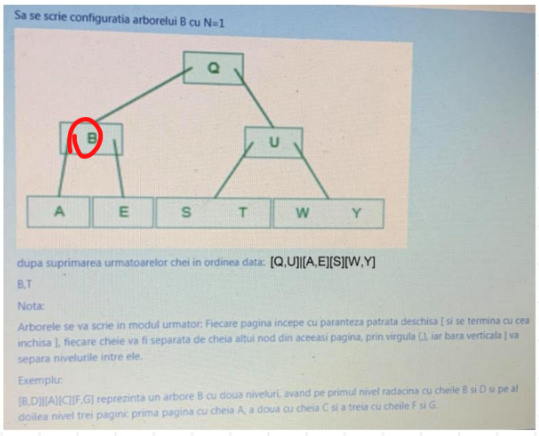


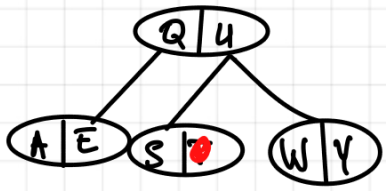


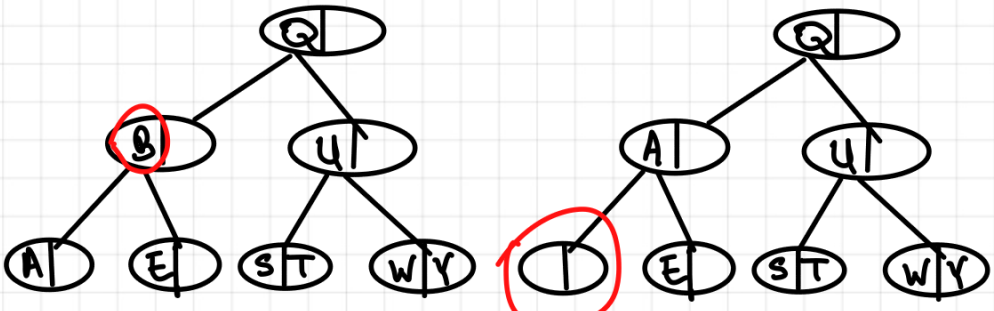




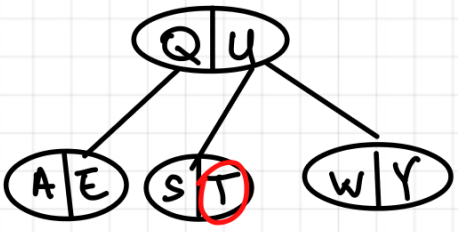
5.

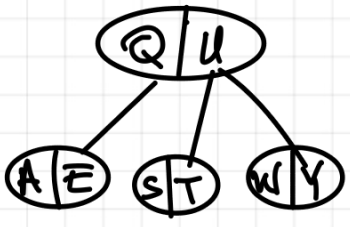

Sa se scrie configuratia arborelui B cu N=1
Q
B
U
A
E
S
T
W
Y
dupa suprimarea urmatoarelor chei in ordinea data: [Q,U][A,E][S][W,Y]
B,T
Nota:
Arborele se va scrie in modul urmator: Fiecare pagina incepe cu paranteza patrata deschisa [si se termina cu cea inchisa], fiecare cheie va fi separata de cheia altui nod din aceeasi pagina, prin virgula (,), iar bara verticala | va separa nivelurile intre ele.
Exemplu:
[B,D][A][C][F,G] reprezinta un arbore B cu doua niveluri, avand pe primul nivel radacina cu cheile B si D si pe al doilea nivel trei pagini: prima pagina cu cheia A, a doua cu cheia C si a treia cu cheile F si G.


Q U
A E
S T
W Y

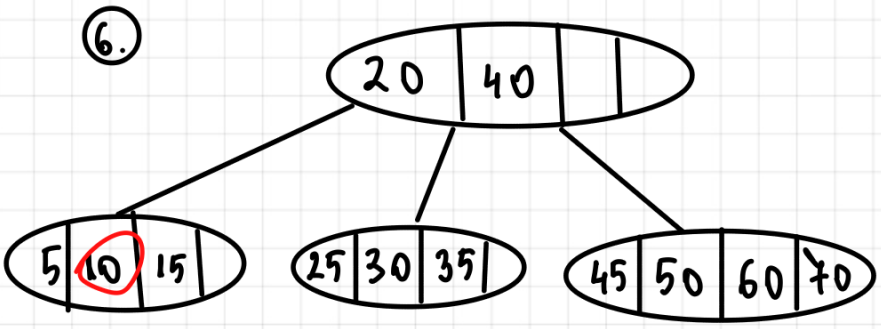

Q
B
U
A
E
S T
W Y

Q
A
U
E
S T
W Y

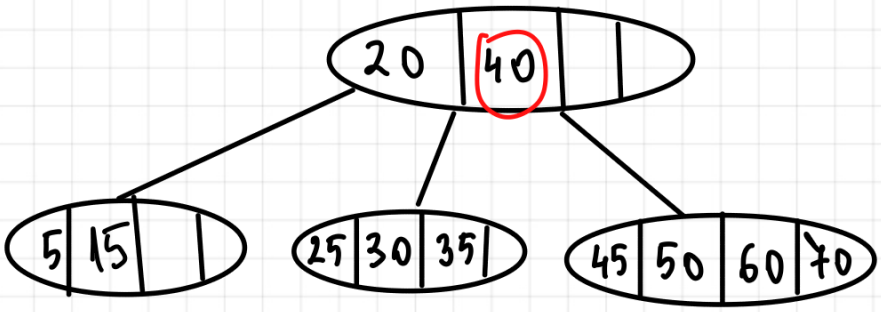

Q U
A E
S T
W Y

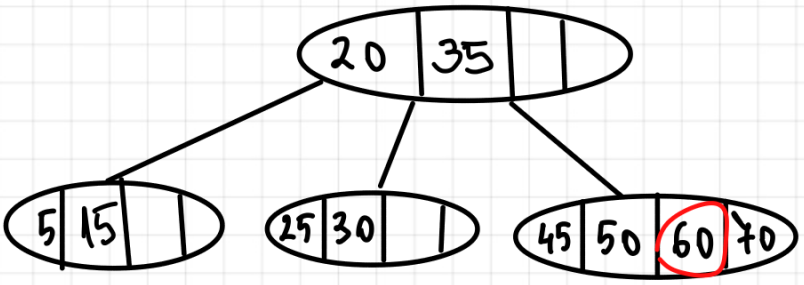

Q U
A E
S T
W Y

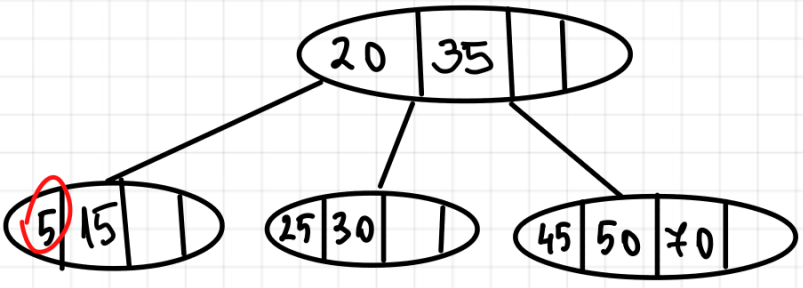
6.

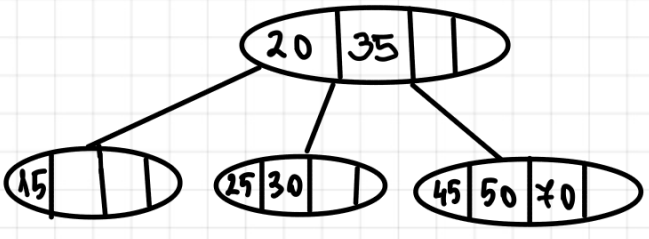


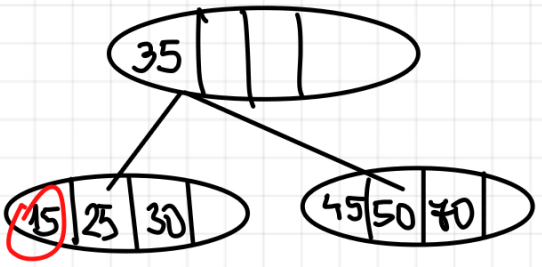
De stals : 10, 40, 60, 5, 15, 30

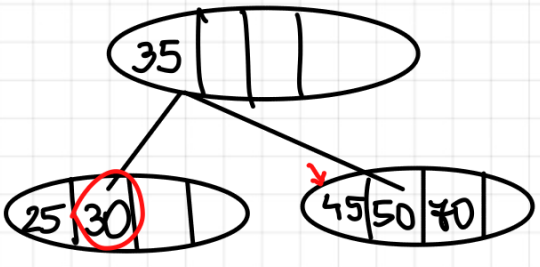


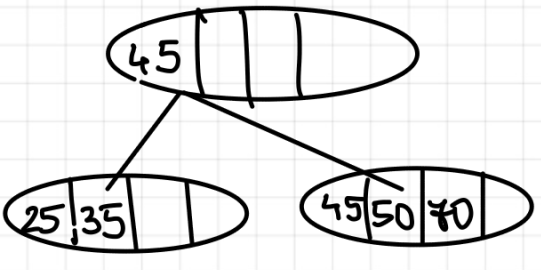


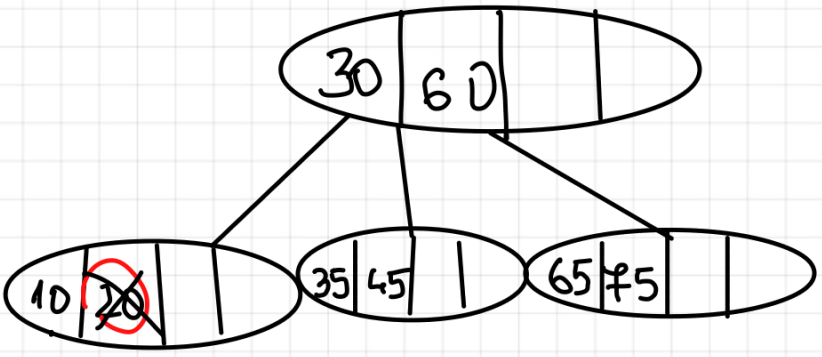




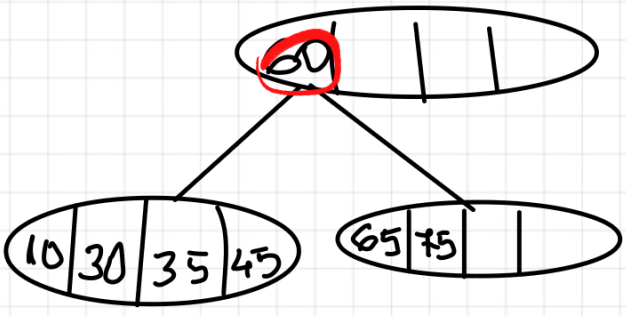


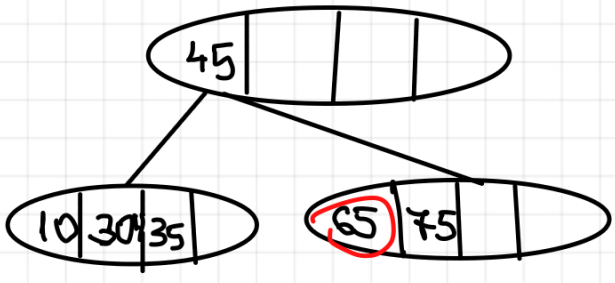


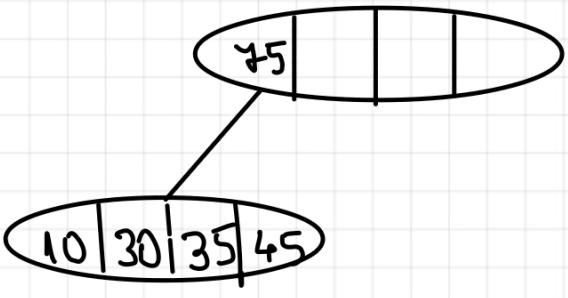


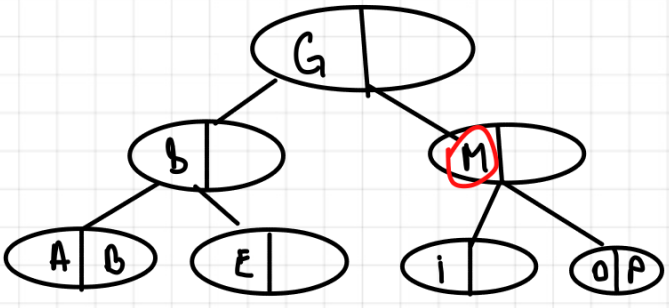

30
60
10
20
35
45
65
75

20, 60, 65

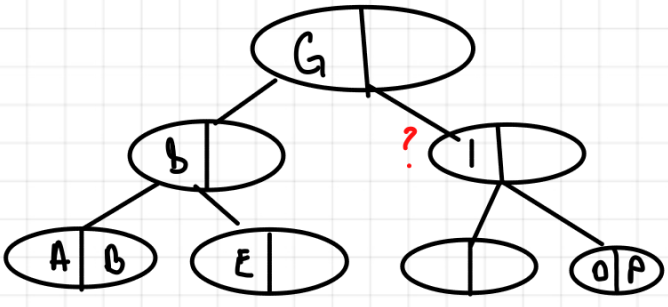

60
10
30
35
45
65
75

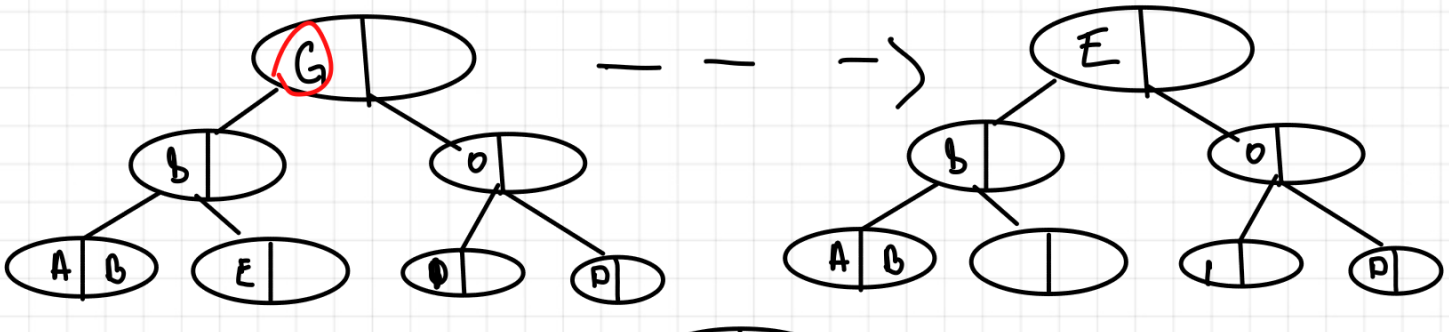

45
10
30
35
65
75

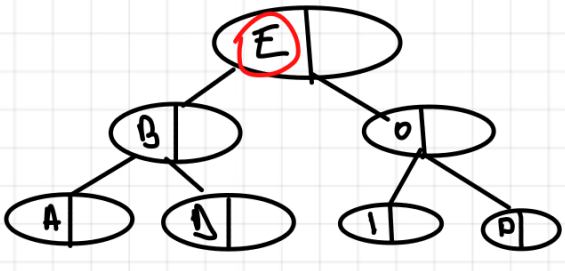

75
10
30
35
45

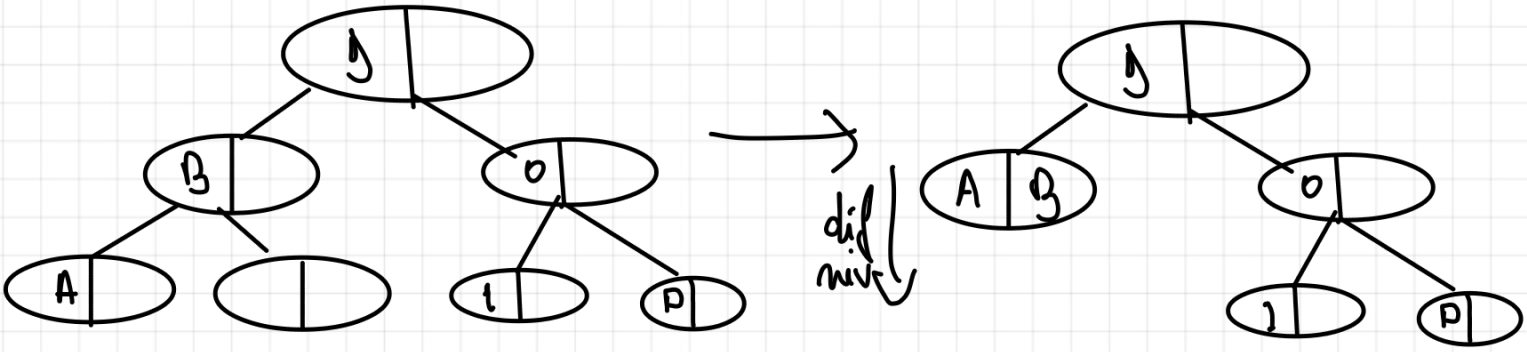

G
b
M
A
B
E
i
O
P

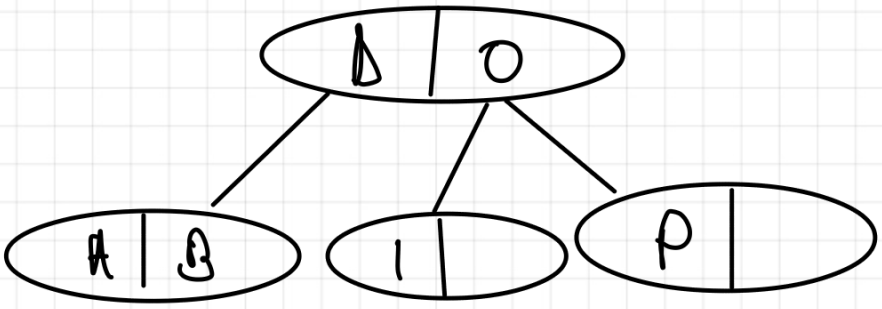
M, G, E

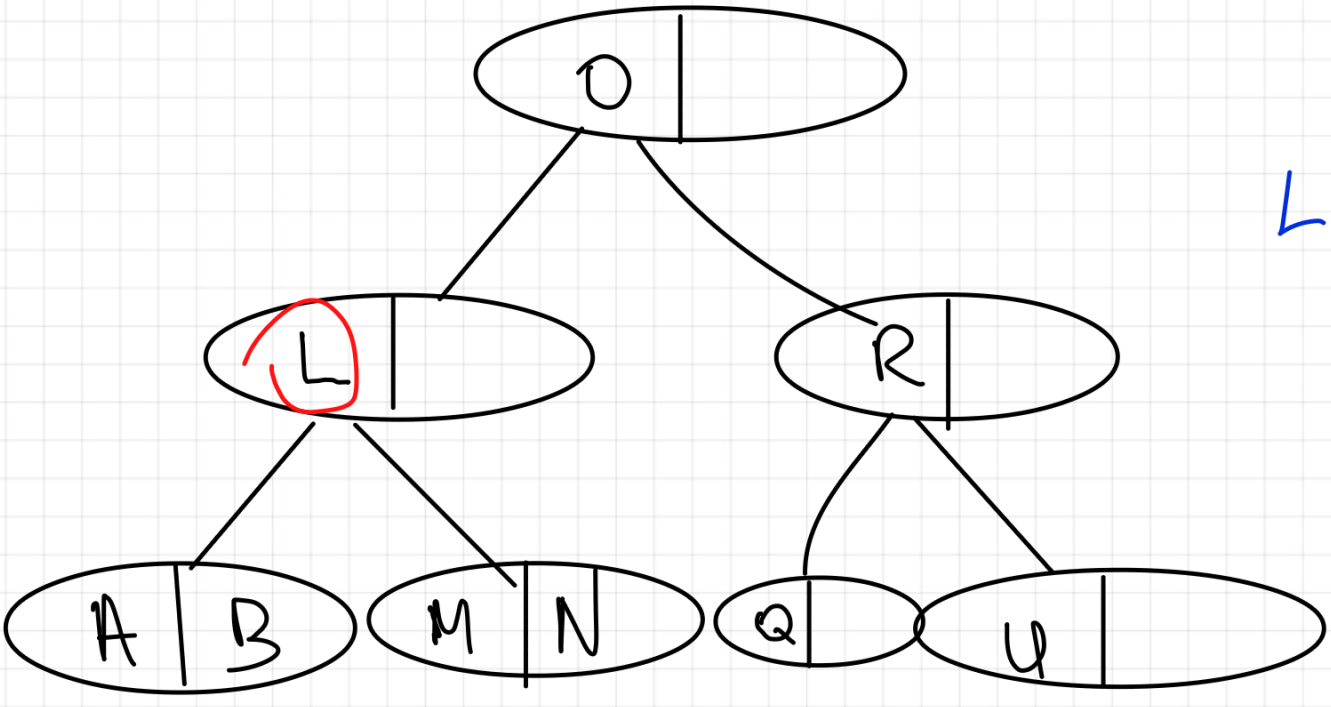

G
b
?
I
A
B
E
O
P


G
b
O
A
B
E
O
P
E
b
O
A
B
I
P

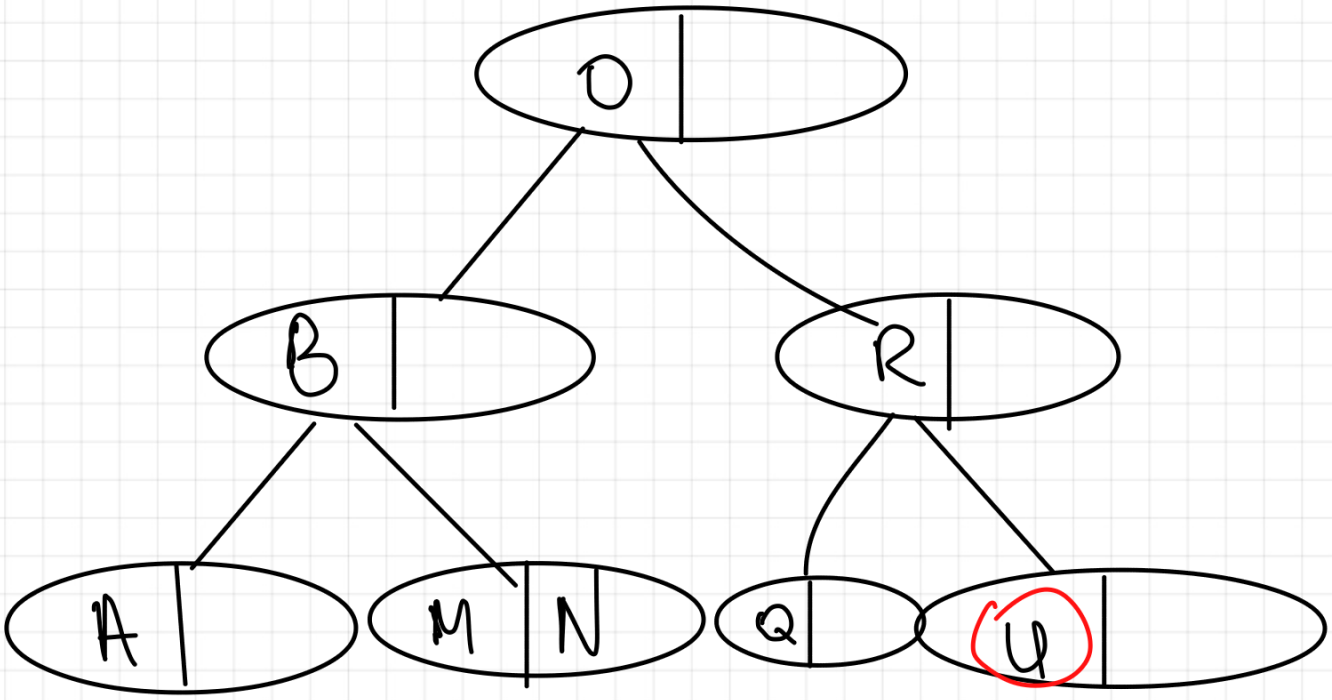

E
B
O
A
I
P

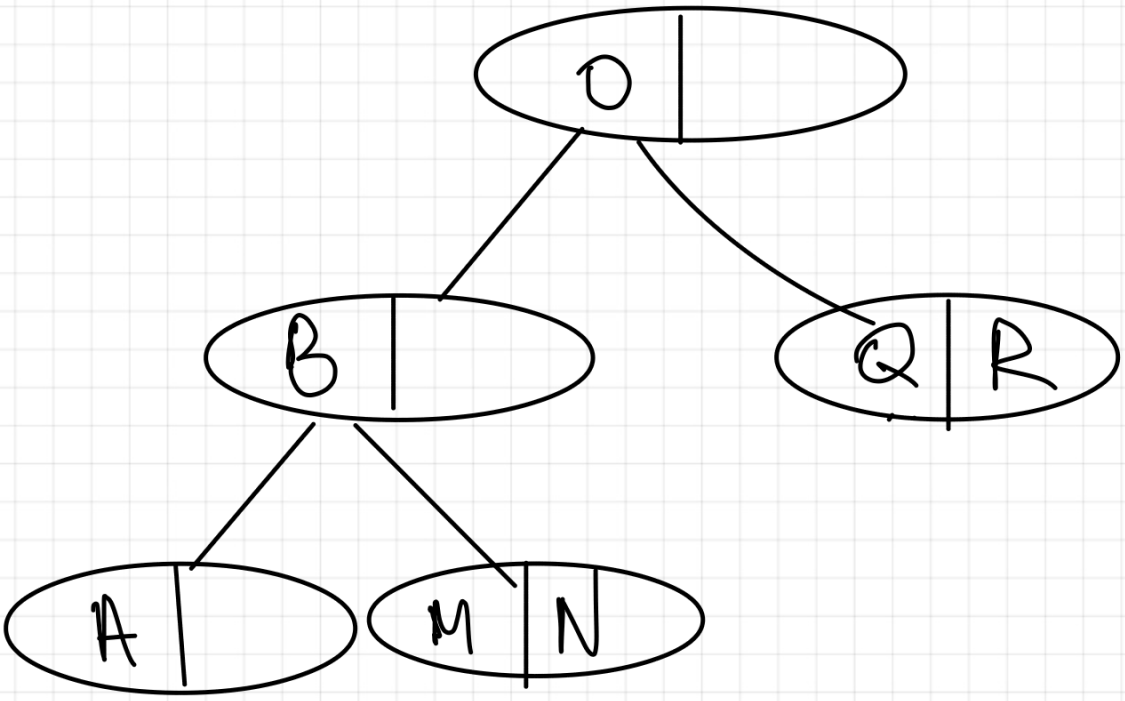






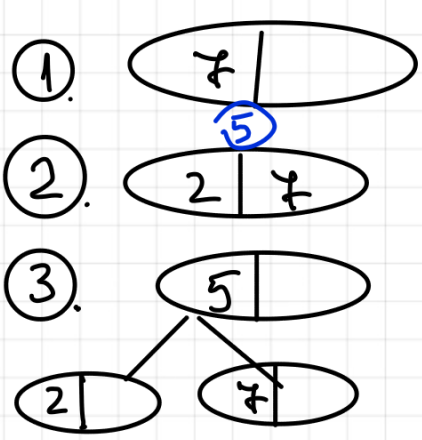
L, 4

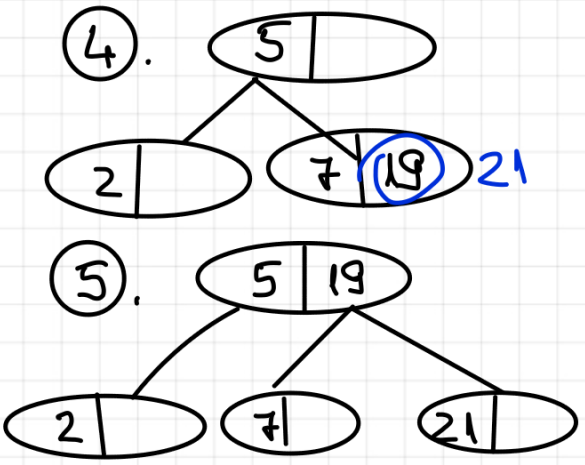


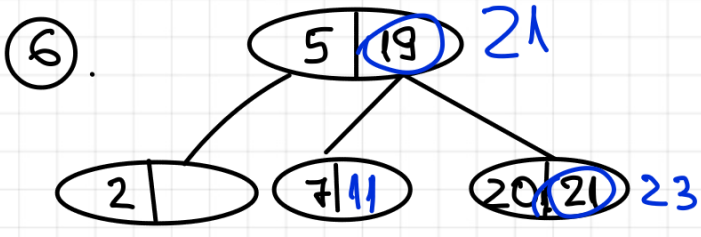


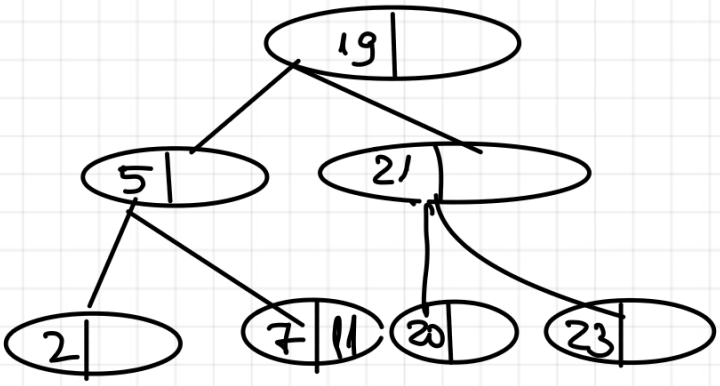
INSERARE

7, 2, 5, 19, 21, 20, 11

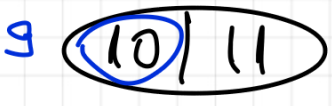


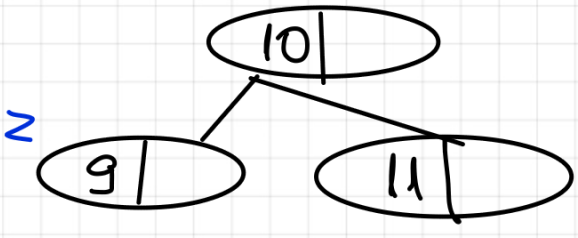


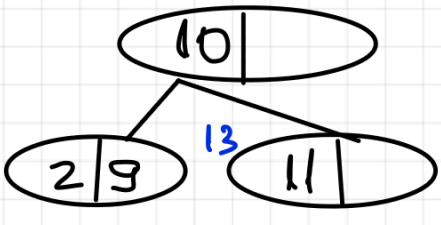


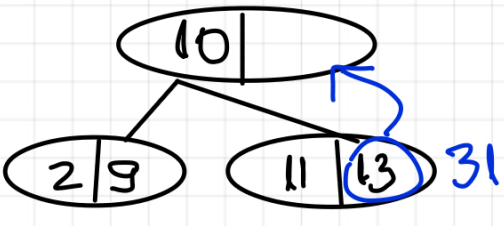


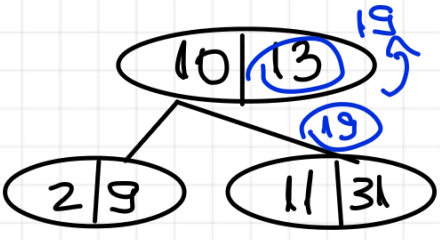
10, 11, 9, 2, 13, 31, 19, 30, 17

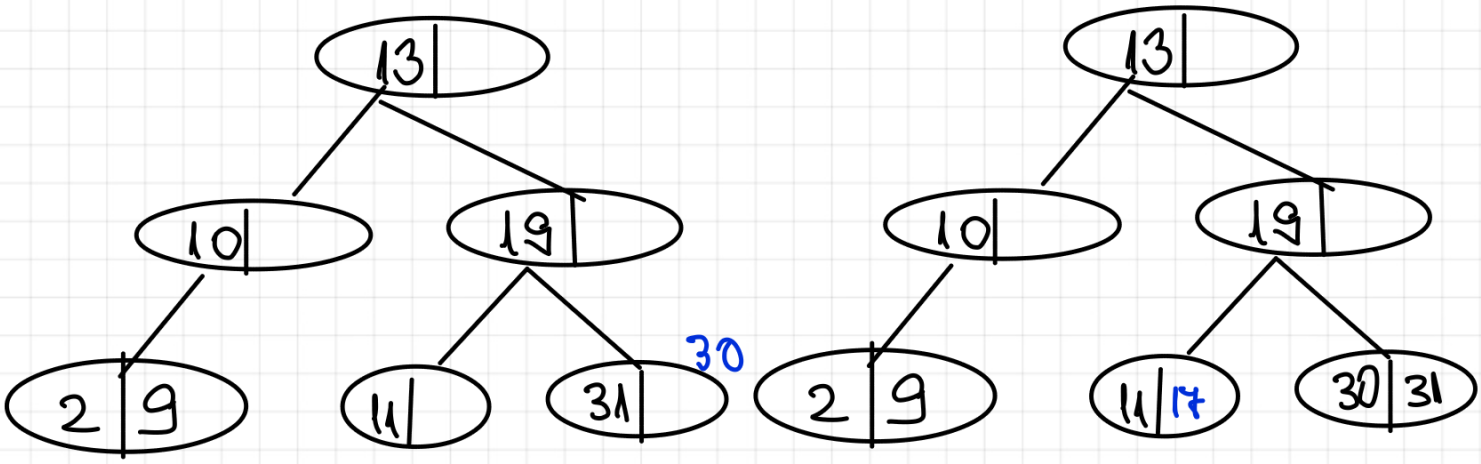




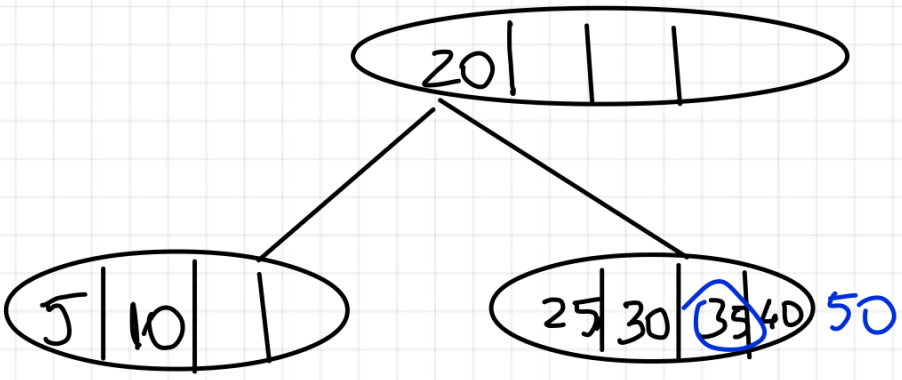


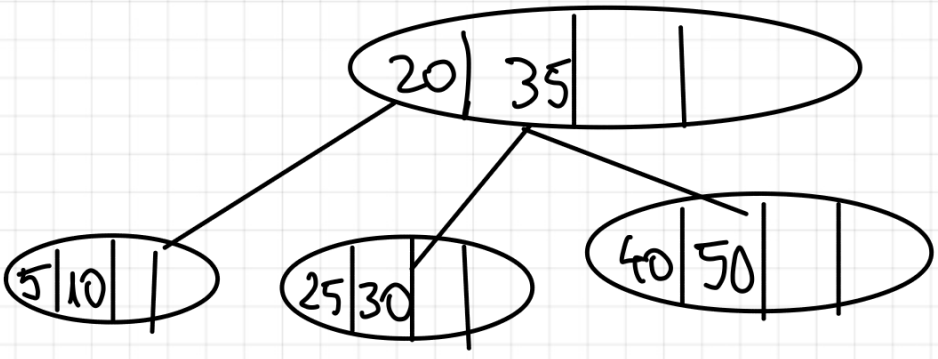




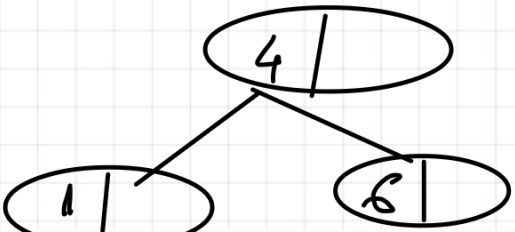


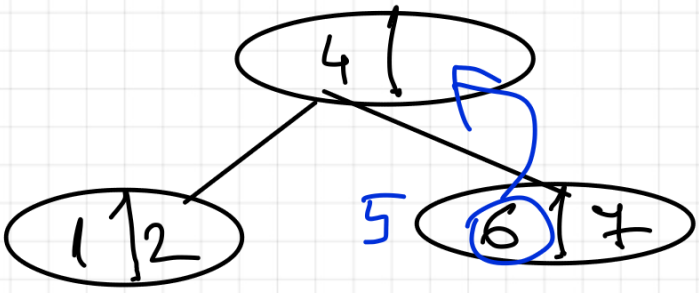
10, 20, 5, 30, 40, 25, 35, 50

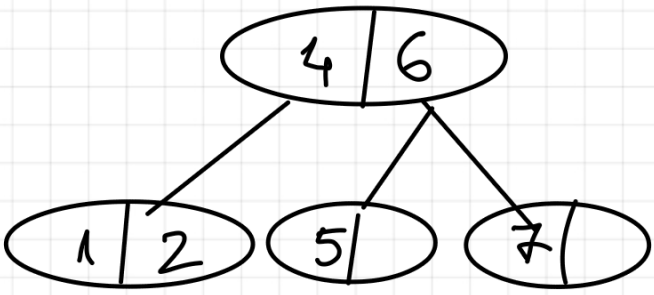


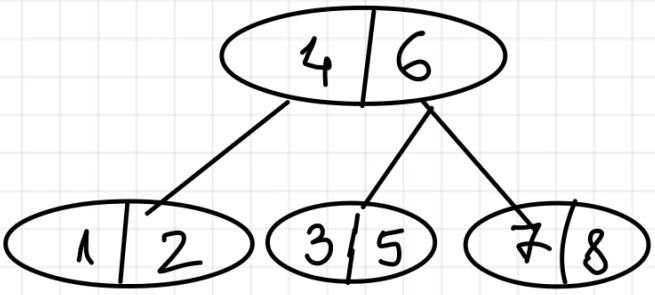


4, 1, 6, 2, 7, 5, 3, 8









50, 15, 70, 35, 10, 60, 5, 30, 20, 40